
MSAlign: Aligning Molecule and Mass Spectra Foundation Models for Metabolite Identification

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Abstract

Accurately identifying metabolites i.e. small molecules from mass spectrometry data remains a core challenge in metabolomics, with broad applications in drug discovery, environmental analysis, and clinical research. We address the Molecule Retrieval task, which consists in recovering the chemical structure of a metabolite from its MS/MS spectrum given a set of candidate molecules. While the recent release of benchmark datasets such as MassSpecGym and Spectraverse has considerably accelerated the development of novel machine learning approaches, the complexity of data preprocessing pipelines and the lack of unified implementations make methods and results difficult to reproduce and compare. We make three contributions. First, we propose a unified framework encompassing recent approaches based on representation alignment and contrastive learning. Second, we introduce MSAlign, inspired by multimodal alignment in vision-language models, which learns a shared representation space by aligning two frozen foundation models (DreaMS for mass spectra and ChemBERTa for molecules) through lightweight MLP projections trained with a candidate-based contrastive objective. MSAlign is simple to implement, fast to train and consistently outperforms existing approaches across all benchmarks. Third, we investigate a long-standing evaluation problem: data splitting strategies in molecule retrieval implicitly trade off data leakage against domain shift. We formalize this tension by introducing a quantitative measure of distribution shift, and use it to evaluate splitting strategies in existing benchmarks. All datasets, splits, candidate sets, and a unified implementation of MSAlign and baselines are publicly released to support reproducible research.

1 Introduction

Metabolomics concerns the study of small molecules in biological systems and plays a central role in understanding disease mechanisms, discovering biomarkers, and enabling therapeutic development [50, 65]. To identify small molecules (metabolites) present in biological samples, biochemists rely mainly on Liquid Chromatography coupled with tandem Mass Spectrometry (LC-MS/MS) [2]. In a nutshell, the different types of ions generated from the molecule entering the mass spectrometer can be selected individually according to their mass-to-charge ratio (m/z) and fragmented at specific

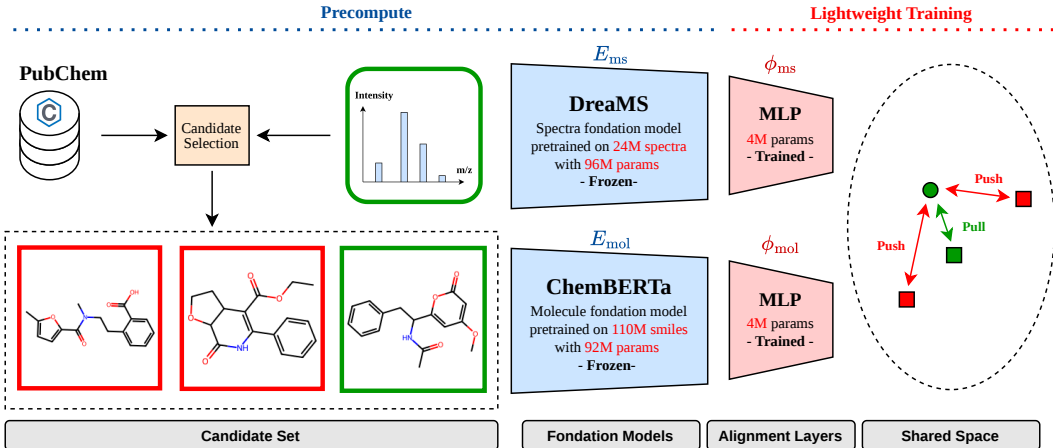


Figure 1: Lightweight alignment of unimodal foundation models via candidate contrastive learning.

collision energies [16]. The resulting measured MS/MS spectrum consists of a list of tuples (m/z , intensity) corresponding to the fragment ions, which provides information about the chemical bonds and substructures of the compound [14]. The central challenge of *Metabolite Identification* is to recover the full 2D chemical structure of a metabolite from its MS/MS spectrum. This task is both analytically critical and, fundamentally hard [20, 60, 68]. The dominant practical approach to this problem relies on spectral library matching, where an experimental spectrum is compared against a database of reference spectra acquired from known compounds and the best-matching entry is reported as the putative identification [38, 57]. While effective when a match exists, this strategy is fundamentally limited by database coverage as, despite significant efforts to increase their size, spectral libraries currently cover less than 1% of the 110 million compounds described in PubMed [4] leaving the vast majority of molecules unidentified.

Machine learning has emerged as a promising paradigm to overcome this coverage bottleneck [30, 46] with rapid progress in recent years, driven by the development of large-scale datasets and the emergence of the first foundation models for mass spectrometry [9]. Molecule identification can be formulated as a supervised prediction problem, where the input is a mass spectrum and the target is a molecular structure. As the full space of metabolites remains unknown, the more general structure elucidation setting, often referred to as *de novo* molecule discovery, is particularly difficult to tackle (see e.g., [56]), and performance remains limited [5, 43, 58, 64].

In this work, we focus on a more tractable variant of the problem, the so-called *Molecule Retrieval* task: given a mass spectrum and a set of candidate molecules, the goal is to retrieve the most likely molecular structure among the candidates. This formulation closely matches practical settings, where spectra are queried against large molecular databases (e.g., PubChem) to identify plausible matches. Most of the proposed ML methods, even if not presented as such, fit the general energy-based framework [42], well suited to tackle structured output prediction as a retrieval task. The approaches to learning the scoring function are divided into two groups: some rely on fingerprint regression [19, 30] or more sophisticated variants [7, 15, 21], while others focus on learning spectra or molecule representations in a contrastive way [11, 27, 36]. In this work, we build on the latter approaches, exploiting the recent advances of multimodal alignment and chemical foundational models [1, 9].

Contributions and paper overview. This paper addresses molecular retrieval from mass spectra through several complementary contributions, spanning methodology, benchmarking, and reproducibility. We introduce MSAAlign, a simple yet effective approach that achieves state-of-the-art performance by aligning off-the-shelf foundation models to learn a shared representation of molecules and mass spectra. Beyond the method itself, we examine the relationships between prior methods and address critical challenges in benchmark design, with the broader goal of advancing the field.

- **Unified framework.** We introduce a unified framework to analyze existing approaches to molecule retrieval, clarifying the relationships between prior methods (Section 2).
- **MSAlign.** We present MSAAlign (Fig. 1), detail the key design choices underlying the method and connections to recent developments in vision-language alignment. (Section 3).

Table 1: Comparison of baselines methods within the proposed unified framework.

Method	Spectra encoder		Molecule encoder		Training objective		Unsupervised data		Require Formula
	Fixed (E_{ms})	Trained (ϕ_{ms})	Fixed (E_{mol})	Trained (ϕ_{mol})	Pretrain	Finetune	Spectra	Mol.	
FFN	Binarize Spectra	—	Fingerprint	—	$\mathcal{L}_{FP}$ (3)	—	×	×	No
DeepSets	Fourier Features	DeepSets	Fingerprint	—	$\mathcal{L}_{FP}$ (3)	—	×	×	No
MIST	Peaks Subformulas	Transformer	Fingerprint	MLP	$\mathcal{L}_{FP}$ (3)	$\mathcal{L}_{cand}$ (5)	Simulated	Negatives	Yes
Emb-Cos	Binarize Spectra	MLP	Fingerprint	MLP	$\mathcal{L}_{cand}$ (5)	—	×	Negatives	No
JESTR	Binarize Spectra	MLP	Graph	GNN	$\mathcal{L}_{batch}$ (4)	$\mathcal{L}_{cand}$ (5)	×	Negatives	No
FLARE	Peaks Subformulas	Transformer	Graph	GNN	$\mathcal{L}_{batch}$ (4)	$\mathcal{L}_{cand}$ (5)	×	Negatives	Yes
MSAlign	DreaMS	MLP	ChemBERTa	MLP	$\mathcal{L}_{cand}$ (5)	—	DreaMS	ChemBERTa	No

- **Distribution shift analysis.** We formalize a core tension in existing benchmarks: splits that minimize data leakage tend to maximize domain shift, and vice versa. We introduce a quantitative metric to navigate this trade-off and improve existing benchmarks interpretability (Section 4).
- **Experiments & ablations.** We provide extensive experimental results demonstrating state-of-the-art performance (Section 5.2) and validating the main design choices (Section 5.3).
- **Open-source codebase.** We will release a unified, open source and fully reproducible implementation of MSAlign and all baselines, enabling the community to build on our work.

2 A unified framework for molecule retrieval

In this section, we introduce a unified framework to analyze the different approaches proposed in the recent literature for molecule retrieval from MS/MS spectra.

Molecule retrieval problem. Let $\mathcal{S}$ denote the space of mass spectra and $\mathcal{M}$ the space of molecules. Given a dataset of paired observations $(s_i, m_i)_{i=1}^N \in \mathcal{S} \times \mathcal{M}$, the goal of Metabolite Identification is to learn a model $f : \mathcal{S} \rightarrow \mathcal{M}$ that predicts the 2D molecular structure associated with a given spectrum. This problem presents difficult challenges due to the discreteness and combinatorial nature of the output space $\mathcal{M}$ and fits the general framework of Structured Output Prediction [3, 47]. As many other discrete prediction tasks, Molecule Identification is generally cast into a Molecule Retrieval problem, where the search is limited to a candidate set $C(s) \subset \mathcal{M}$. In that case, f_θ , the retrieval function, based on a scoring function $\rho_\theta : \mathcal{S} \times \mathcal{M} \rightarrow \mathbb{R}$ of parameter θ , writes as follows:

$$f_\theta(s) = \arg \max_{m' \in C(s)} \rho_\theta(s, m'). \quad (1)$$

Then, the problem reduces to learning a scoring function that assign higher scores to the correct pairs than to the incorrect ones which connects to the large body of works in energy-based models [42].

In our problem, the candidate sets are typically defined via a mass filter, i.e., $C(s) = \{m' \in \mathcal{M} : \text{abs}(|m'| - |m_s|) < \epsilon\}$ where $|m'|$ is the mass of any known molecule, ϵ is a small threshold, and $|m_s|$ is the mass of the true molecule, which is assumed to be known. This is a reasonable assumption, as the determination of the mass of the neutral molecule given the MS/MS spectrum (i.e., the determination of the ion species that was fragmented) is a well-managed task both manually and automatically [60].

Scoring function and training loss. We focus here on the large class of scoring functions ρ_θ defined as a similarity $\text{sim} : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}^+$ over a shared representation $\mathcal{Z}$ of spectra and molecules, i.e.

$$\rho_\theta(s, m) = \text{sim}(g_\theta(s), h_\theta(m)) \quad (2)$$

where $g_\theta : \mathcal{S} \rightarrow \mathcal{Z}$ and $h_\theta : \mathcal{M} \rightarrow \mathcal{Z}$ are encoders or *feature maps* and sim is typically the cosine similarity. The following approaches have been proposed to learn the score:

Retrieval through regression. One way to learn the function ρ_θ is to concentrate on learning the feature map g_θ while keeping $h_\theta(m) = h(m)$ fixed, either as a molecular fingerprint $FP(m)$ i.e. a binary vector representing the presence/absence of certain substructures [26, 30, 34] or as a feature vector in a Hilbert space [6, 7]. Learning the score function boils down to estimate the regression of the variable $h(m)$ by the variable s . Taking the example of fingerprint regression, the loss function

can then be chosen as:

$$\mathcal{L}_{\text{FP}}(\theta) = - \sum_{i=1}^N \text{sim}(g_{\theta}(s_i), \text{FP}(m_i)). \quad (3)$$

Retrieval through alignment. An alternative to regression is to jointly learn the feature maps g_{θ} and h_{θ} . This formulation connects self-supervised multimodal alignment [51] with supervised retrieval, naturally leading to contrastive learning objectives [10, 63]. The most common choice is the InfoNCE loss [11, 15, 36], which encourages the similarity of the true pair (s, m) to be higher than that of negative pairs (s, m') . Several variants exist depending on how negatives are selected.

As training is performed through mini-batch iteration, a common choice is to use the other molecules in the batch as negatives, leading to the following *in-batch* InfoNCE loss:

$$\mathcal{L}_{\text{batch}}(\theta) = - \sum_{i=1}^B \log \frac{\exp(\rho_{\theta}(s_i, m_i))}{\sum_{j=1}^B \exp(\rho_{\theta}(s_i, m_j))}. \quad (4)$$

This is particularly computationally efficient as the computation of all pairwise similarities $\rho_{\theta}(s_i, m_j)$ requires only to encode B spectra ($g_{\theta}(s_i)$) and B molecules ($h_{\theta}(m_j)$). However, this approach is limited by the fact that in-batch negatives are not necessarily hard to distinguish, making the training loss less informative. A large body of work attempt to address this limitation through hard negative mining or by scaling the number of negatives [53, 59, 69].

In molecule retrieval, an additional structure can be exploited: for each spectrum s_i , a candidate set $C(s_i)$ provides a natural source of hard negatives. This leads to the *candidate-based* InfoNCE loss:

$$\mathcal{L}_{\text{cand}}(\theta) = - \sum_{i=1}^B \log \frac{\exp(\rho_{\theta}(s_i, m_i))}{\sum_{m' \in C_K(s_i)} \exp(\rho_{\theta}(s_i, m'))}, \quad (5)$$

where $C_K(s_i) \subset C(s_i)$ is a subset of K negatives sampled from the candidate set. While more aligned with the retrieval task, this loss is computationally expensive as it requires $K \times B$ similarity evaluations with limited reuse of molecule embeddings across spectra.

Critically, our approach enables the use of (5) at low computational cost. In Section 5.3, we provide a systematic comparison of regression, in-batch, and candidate-based losses, showing that hard negatives are critical for strong performance in metabolite identification.

Architecture and data representation. The second key design choice is the architecture of the embeddings g_{θ} and h_{θ} . While a wide range of representations have been explored in the recent literature, we focus on representations in $\mathcal{Z} \subset \mathbb{R}^d$ and refer to [49] for a comprehensive overview.

For spectra, common approaches include discretizing the m/z axis into fixed bins [15, 36], applying convolutional neural networks [40], or modeling spectra as sequences of peaks (optionally enriched with formula annotations) and processed by transformer architectures [11, 13, 26].

For molecules, representations range from fixed fingerprints to learned embeddings. Classical approaches rely on handcrafted fingerprints, optionally refined with an MLP [26], while more recent methods use graph neural networks (GNNs) to directly encode molecular structures [11, 36]. An alternative paradigm represents molecules as SMILES or SELFIES strings and leverages transformer-based language models to learn molecular embeddings [22, 29].

Unified framework. We summarize the possible encoding choices in a compact formulation:

$$g_{\theta_1}(s) = (\phi_{ms}^{\theta_1} \circ E_{ms})(s), \quad (6)$$

$$h_{\theta_2}(m) = (\phi_{mol}^{\theta_2} \circ E_{mol})(m), \quad (7)$$

in which E_{ms} and E_{mol} are non-trainable preprocessing steps (e.g. binning for spectra, fingerprinting for molecules) and $\phi_{ms}^{\theta_1}$ and $\phi_{mol}^{\theta_2}$ are trainable encoders (e.g. MLP, GNN, transformer). Under this formulation, a large body of work can be reduced to the choice of E_{ms} , E_{mol} , $\phi_{ms}^{\theta_1}$ and $\phi_{mol}^{\theta_2}$, as well as the loss function among those presented in (3), (4) or (5). Table 1 summarizes the comparison between recent methods along with the key design features discussed above.

3 Proposed Method: MSAlign

We now present MSAlign, the intuition behind its design, its adequation to the general framework introduced in Section 2, and the key design choices contributing to its strong performance, all of which are further analyzed in ablation studies 5.3.

Motivation. MSAlign is inspired by recent progress in vision–language learning, where aligning pretrained unimodal models has been shown to be competitive with joint multimodal pretraining while being more computationally efficient [62, 69]. This strategy is particularly well suited to settings such as metabolite identification, where paired data are limited but large amounts of unimodal data are readily available [55]. In short, we propose to align pretrained foundation models for spectra and molecules instead of training a multimodal model from scratch.

Foundation models for MS/MS data. On the spectral side, we use DreaMS [9], the first foundation model for mass spectra, pretrained on 24 million high-quality MS/MS spectra. On the molecular side, a large body of work has explored graph-based and text-based foundation models such as UniMol2 [35], MolE [45], ChemBerta [1], Grover [54] and GRALE [39]. ChemBERTa appears as a natural choice, both because it is simple to use and because it achieves strong performance on MoleculeNet benchmarks [66]. Table 10 further shows that it outperforms other comparable molecular foundation models, including Grover, SMITED, and MHG-GED when used in the MSAlign framework.

Architecture. MSAlign fits the framework introduced in Section 2. In MSAlign, E_{ms} and E_{mol} are the foundation models DreaMS and ChemBERTa, respectively, and ϕ_{ms}^θ and ϕ_{mol}^θ are lightweight MLP projection layers that map both modalities in a shared space. This design has two main advantages. First, it leverages strong pretrained representations, outperforming classical vectorization strategies such as binned spectra and fingerprints, as shown in Tables 5 and 11. Second, it remains lightweight, as the pretrained encoders are kept frozen and only small projection layers are trained (about 4M parameters, compared to 96M and 92M for DreaMS and ChemBERTa, respectively).

Training. The final key design choice is the training objective. Table 4, Figure 3 and Figure 5 show that candidate-based InfoNCE (5) consistently outperforms both regression-based approaches and in-batch contrastive learning, highlighting the importance of hard negatives for molecule identification.

This formulation may seem computationally prohibitive, but MSAlign makes it tractable in practice. The low-dimensional embeddings produced by DreaMS and ChemBERTa (dimension 1024 and 768, respectively) can be precomputed and efficiently stored, enabling fast similarity computation over candidate sets. As a result, MSAlign can leverage candidate-based InfoNCE while maintaining lightspeed training times (about 20 minutes on a single V100 for MassSpecGym with $K = 64$).

4 Assessing the challenges of data splitting in metabolite identification

Before moving to the experimental evaluation of MSAlign, we first discuss a critical aspect of the experimental protocol: the choice of data splitting strategy.

Challenge. Data splitting is a non-trivial issue for MS/MS datasets. Because the same molecule often appears across multiple spectra under different ionization conditions, naive splits are highly susceptible to data leakage. In contrast, enforcing strong dissimilarity between training and test sets can induce significant distribution shifts [52], which may not reflect practical deployment settings and would require dedicated domain adaptation techniques [24].

To help the community navigate this trade-off between data leakage and distribution shift, we now propose a quantitative metric to evaluate the degree of shift induced by a given splitting strategy.

Quantitative evaluation of domain shift. A natural approach to quantify the shift between two distributions is to consider the Wasserstein distance $W(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}})$ [48]. Unfortunately, its raw value is difficult to interpret, as it depends on the scale of the underlying embedding space. To address this issue, we propose to normalize it by an estimate of the intrinsic variability of the training data.

Specifically, we randomly partition $\mathcal{D}_{\text{train}}$ into two disjoint subsets $\mathcal{D}_{\text{train}} = \mathcal{D}_{\text{train}}^1 \cup \mathcal{D}_{\text{train}}^2$, and define:

$$\text{Shift}(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) = \frac{W(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}})}{W(\mathcal{D}_{\text{train}}^1, \mathcal{D}_{\text{train}}^2)}. \quad (8)$$

A value close to 1 indicates a test distribution comparable to within-train variability, while larger values reflect stronger distribution shifts. The Wasserstein distance can only be computed for distributions defined over a common embedding space. To this end, we embed each spectrum–molecule pair as $E(s, m) = [E_{\text{ms}}(s), E_{\text{mol}}(m)]$ using DreaMS and ChemBERTa. In the following, we approximate the Wasserstein distance using the sliced Wasserstein with $p = 100$ projections and 5 random seeds, implemented in the POT toolbox [25].

Table 2: For different splitting strategies of MassSpecGym, we report the Wasserstein train/test distribution shift from equation (8), along with MSAAlign test performances.

Split	Shift	R@1	R@5	R@20
MCES	4.18 ± 0.27	16.24	35.66	59.97
Murcko	4.01 ± 0.09	24.75	47.89	72.57
Formula	1.40 ± 0.10	53.83	73.11	87.10
2D InChI	1.29 ± 0.06	54.34	74.54	87.13
Random	0.70 ± 0.01	93.21	97.55	99.23

Note that we consider the joint distribution of spectra and molecules, hence the proposed metric account both for shifts in the spectra and in the molecules.

Possible strategies. A variety of splitting strategies have been proposed in the literature, most aiming to ensure that test molecules differ sufficiently from those in the training set, with various definitions of “dissimilarity”. The MCES split of MassSpecGym [52] is based on the Maximum Common Edge Subgraph (MCES) similarity and enforce enforces a minimum MCES distance > 10 via clustering. The Murcko scaffold split [9] coarsely groups molecules by their core scaffold structure. We also consider the Formula split, which enforces distinct chemical formulas across splits; and the 2D InChIKey split based on the 14 first digits of the International Chemical Identifier hash, a classical weaker constraint allowing identical formulas but different 2D structures [19]. As a reference, we also consider an unconstrained random split.

Analysis on MassSpecGym. We now evaluate these strategies on MassSpecGym by reporting both the distribution shift (Eq. 8) and the test retrieval performance of MSAAlign (Table 2).

First of all, we observe that MCES and Murcko splits induce large shifts ($\text{Shift} > 4$). In figure 2, we show that the MCES split induces a particularly strong shift, with the test set forming a distinct cluster far from the training data (see Figure 6 to compare with other splits). Hence, those splits are particularly challenging, and can be used to measure the out-of-distribution generalization capabilities of a model, this may be relevant for specific applications such as de novo drug discovery. However, such a strong shift may not reflect practical deployment in retrieval settings, where test molecules are likely to share some similarity with those in the training set. We encourage future works to address these challenging splits with dedicated domain adaptation techniques.

On the opposite end of the spectrum, the random split induces a very small shift ($\text{Shift} < 1$), leading to near-perfect performance. However, this is clearly an unrealistic setting, as it allows for significant data leakage, with the same molecule appearing in both training and test sets under different ionization conditions.

Overall, the Formula and 2D InChIKey splits provide a more balanced trade-off, with Shift values around 1-2. We conclude that Formula-split is sufficient to prevent data leakage while still reflecting

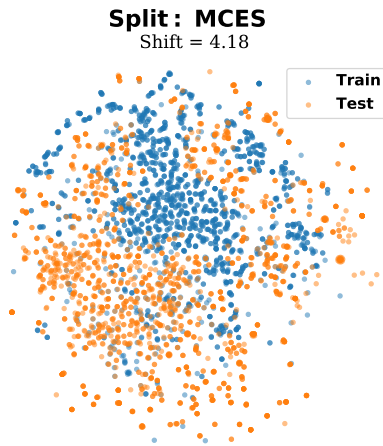


Figure 2: t-SNE visualization of the MassSpecGym MCES splits.

Table 3: Retrieval performances of MSAlign and the baselines. Best results in **bold**, second-best is underlined. All values are percentages.

Dataset	Metric	Models that don’t require formula						If Formula is known		
		FFN	DeepSets	JESTR	Emb-Cos	SAIL	MSAlign	MIST	FLARE	MSAlign (+Filter)
NPLIB1	R@1	8.9	11.1	11.2	<u>22.5</u>	18.6	31.8	27.2	19.6	34.7
	R@5	22.0	30.4	33.1	<u>48.3</u>	46.4	60.4	<u>52.0</u>	50.8	64.7
	R@20	44.4	56.5	62.0	<u>72.7</u>	<u>75.8</u>	82.9	73.8	<u>76.6</u>	87.9
Spectraverse	R@1	9.5	12.7	11.3	27.1	14.5	32.3	<u>20.2</u>	12.3	39.4
	R@5	18.9	24.4	32.9	<u>51.8</u>	34.3	59.1	<u>40.9</u>	33.7	69.0
	R@20	34.7	44.3	60.1	<u>72.2</u>	57.4	79.6	<u>63.8</u>	62.5	87.1
MassSpecGym (Formula Split)	R@1	16.9	22.2	20.0	41.2	29.5	53.8	<u>46.8</u>	31.8	61.3
	R@5	27.8	36.1	40.5	<u>63.4</u>	51.3	73.1	<u>66.1</u>	64.5	82.0
	R@20	40.4	50.9	60.9	<u>79.6</u>	73.0	87.1	79.3	<u>84.1</u>	93.0
MassSpecGym (MCES Split)	R@1	2.54	5.24	11.8	<u>12.3</u>	09.5	16.2	14.6	<u>27.2</u>	32.3
	R@5	7.59	12.58	25.3	<u>26.2</u>	23.9	35.6	34.9	<u>53.8</u>	61.8
	R@20	20.00	28.21	<u>49.7</u>	46.4	45.2	59.9	59.1	<u>80.2</u>	85.5

a realistic deployment setting, and we adopt it for all subsequent experiments. For completeness, we also report results on the more challenging original MCES split of MassSpecGym.

5 Experimental Results

In this section, we first present the experimental setting 5.1, then we report the retrieval performances of MSAlign compared to a range of baselines 5.2. Finally, we validate step by step the key design choices of our method 5.3.

5.1 Experimental Setting

Supervised Datasets We consider three open-source datasets of increasing scale and diversity. **NPLIB1** [18], is the smallest benchmark (10,633 spectra/molecule pairs). **MassSpecGym** [8] provides 231,104 carefully curated and standardized pairs, and is rapidly becoming the community standard for benchmarking. **Spectraverse** [28] is the largest collection (488,797 pairs) and provides broader coverage of adducts and ionization modes, with more 50% of spectra corresponding to non-[M+H]⁺ adducts, making it the most heterogeneous and challenging setting. Note that pair counts overestimate chemical diversity as molecules frequently appear under multiple ionization conditions; Table 12 reports unique molecule counts along with pair counts for all datasets.

Candidates Sets. Following the standardized protocol of MassSpecGym [8], candidate molecules are retrieved from PubChem [37] (118M compounds) by selecting structures whose molecular mass matches the target molecule within a 10 ppm tolerance, prioritizing compounds sampled from databases of decreasing metabolite relevance (see Algorithm 1 of [8] for details). We keep $K = 256$ candidates per spectrum; as illustrated in Figure 4, these candidates are typically structurally close to the target, making retrieval challenging even at this moderate pool size.

Baselines. We propose a comprehensive list of recent baselines : **FFN** and **DeepSets** are fingerprint prediction baselines introduced in MassSpecGym [8]. For DeepSets, we use the more expressive Fourier Features variant, and we keep the original hyperparameters except for the number of epochs, increased from 5 to 50, which significantly improves this baseline. **MIST** [27] is a complex model that pretrains with fingerprint prediction, finetunes with contrastive learning, and augments the data with a forward model for spectra simulation. **JESTR** [36] is a simple contrastive model that pretrains with batch InfoNCE (4) and finetunes with the more computationally expensive candidates InfoNCE (5). **FLARE** [11] is a direct follow-up work that considers a stronger spectra encoder leveraging peak annotation with subformulas and a more interpretable, richer similarity. **Emb-Cos** is the strongest of the four models proposed in [15]; similarly to MSAlign, it uses candidates InfoNCE to learn a shared space for molecules and mass spectra, but it does not use pretrained encoders, which limits expressivity and makes the model approximately 10 times slower to train. **SAIL** was originally

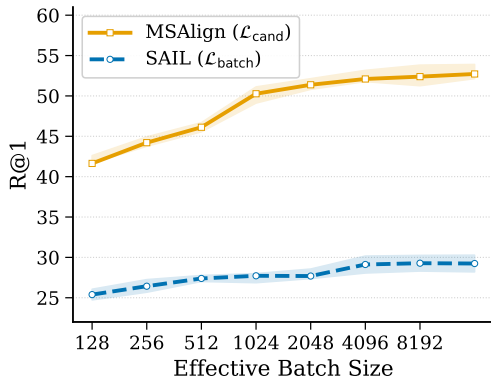


Figure 3: Performance as a function of effective batch size (MassSpecGym). For $\mathcal{L}_{\text{candidate}}$ (MSAlign) the effective batch size is $B \times K$; we fix $B = 128$ and scale the number of negatives K . For $\mathcal{L}_{\text{batch}}$ (SAIL), we scale the batch size B .

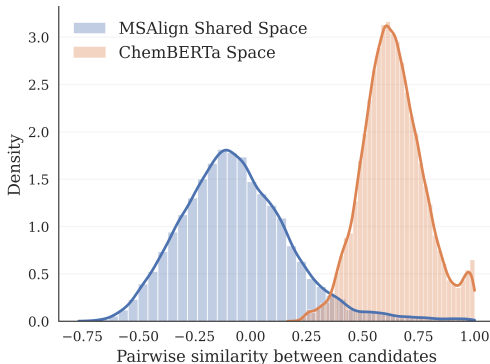


Figure 4: Similarities between pairs of candidates. In the ChemBERTa space, candidates are close to each other (average is $\mu_{pc} = 0.63$), whereas MSAlign learns a space where they are easier to distinguish ($\mu_{pc} = -0.06$).

proposed to align pretrained vision and language models [69], but it can be directly adapted to our setting for aligning molecules (ChemBERTa) and mass spectra (DreaMS); following the original work, we consider a very large batch size for the contrastive loss ($B = 32k$).

Reproducibility. For fair and reproducible results we re-implement all methods within our open source code base. The only exception to this is MIST which is a complex framework that proves challenging to re-implement from scratch [31].

5.2 Retrieval Performances and comparison to competitors

MSAlign and all baselines are trained and evaluated on three publicly available benchmarks (Table 3). MSAlign achieves R@1 scores of 16.2% and 53.8% on MassSpecGym MCES and Formula splits, respectively, and 31.8% and 32.3% on NPLIB1 and Spectraverse.

We note some discrepancies with previously reported results: JESTR performs slightly below its original report, which echoes issues discussed in public implementations¹. We observe a similar trend for FLARE, potentially due to shared implementation components. In contrast, simpler baselines (FFN, DeepSets) achieve stronger results when trained with more epochs, as described in the baseline section. Note that the results we report for NPLIB1 are based on the set of 256 candidates described in 5.1 and cannot be compared to retrieval performance for different candidate sets.

Overall, MSAlign consistently outperforms all baselines across datasets and metrics. These results highlight the benefit of leveraging foundation models, effectively exploiting large-scale unimodal data. They also emphasize the importance of candidate-based contrastive learning as Emb-Cos (that also leverages the same $\mathcal{L}_{\text{cand}}$ objective) emerges as the second strongest baseline.

Finally, some methods (e.g., MIST and FLARE) assume access to the molecular formula. In practice, this information is unknown and must be estimated, which remains an active research problem [17, 32, 67]. For fair comparison, we report their results separately. We additionally evaluate MSAlign under this assumption by filtering candidates to those matching the ground-truth formula, reducing the candidate set to approximately $K \approx 100$. This variant (MSAlign+Filter) outperforms both MIST and FLARE, suggesting that constraining the candidate space via formula information can yield larger gains than subformula-based peak annotation alone.

Table 4: Comparison of training losses. We report the retrieval performances on MassSpecGym (Formula-based split). The regression losses are evaluated in both directions, i.e. mapping spectra to molecules and vice versa.

Training Loss	MassSpecGym		
	R@1	R@5	R@20
Regression (mol $\rightarrow$ ms)	24.0	42.8	59.2
Regression (ms $\rightarrow$ mol)	27.1	47.5	66.0
Contrastive (in-batch)	29.5	51.3	73.0
Contrastive (candidate-based)	53.83	73.11	87.10

Table 5: Comparison of different encoders E_{ms} and E_{mol} . Dimensions of the embeddings are indicated in parentheses. Results are reported on MassSpecGym (Formula-based split). See Table 10 for alternative molecular foundation models.

Encoders		MassSpecGym		
Spectra (dim)	Molecules (dim)	R@1	R@5	R@20
Binned (10500)	Fingerprint (4096)	41.2	63.4	80.0
Binned (10500)	ChemBERTa (768)	45.8	65.9	81.4
DreaMS (1024)	Fingerprint (4096)	48.5	70.1	84.0
DreaMS (1024)	ChemBERTa (768)	53.8	73.1	87.1

5.3 Validating MSAAlign design choices

In the following we analyze the impact of the key design choices of MSAAlign, namely the training loss and the encoders for spectra and molecules. Results are reported on MassSpecGym (Formula-based split), but we show in Appendix C that our conclusions generalize to other datasets.

Training loss. The first major design choice is the training loss. Fixing the other components, we now compare the contrastive objectives (in-batch (4) and candidate-based (5) variants), and regression-based approaches that directly learn a map from spectra to molecules, or vice versa.

Table 4 shows that regression-based approaches perform significantly worse than contrastive ones, especially the mol $\rightarrow$ ms direction, which is expected as the mapping from molecules to spectra is inherently one-to-many. The ms $\rightarrow$ mol direction also underperforms, suggesting that direct regression in the molecular space remains challenging. We hypothesize that this limitation arises from the structure of the molecular embedding space, where candidate molecules tend to be close to each other and thus difficult to distinguish. To test this, we compute the average pairwise similarity between candidates (Figure 4), obtaining $\mu_{pc} = 0.63$ in the ChemBERTa space versus $\mu_{pc} = -0.06$ in the learned shared space. This indicates that MSAAlign learns representations in which candidates are more separable.

We next compare the two contrastive formulations and observe that the candidate-based consistently outperforms the in-batch variant (Table 4). Since the candidate-based InfoNCE is more computationally expensive, we also compare both approaches as a function of their *effective batch size*, defined as B for the in-batch variant and $B \times K$ for the candidate-based variant in Figure 3. Overall, both methods benefit from scale, but the candidate-based formulation remains consistently superior. On MassSpecGym (Formula Split), it scales from 45% R@1 at an effective batch size of $B \times K = 256$ to 53.8% at $B \times K = 16k$. On the other hand, the in-batch variant reaches a plateau around 30% even when scaled to $B = 16k$. In Appendix C, we show that this observation generalizes to other datasets.

Molecule and spectra encoding. The second key design choice in MSAAlign is the selection of encoders for spectra and molecules. To the best of our knowledge, MSAAlign is the first approach to leverage pretrained foundation models for both modalities. We evaluate this choice by comparing MSAAlign to variants using non-trainable encoders, namely Morgan fingerprints for molecules ($d = 4096$) and binned representations for spectra (width 0.1 Da), which are commonly used in prior work [9, 23]. Results in Table 5 show that both pretrained encoders contribute to performance, with DreaMS providing the largest gain. In Appendix C, we show that this observation generalizes to other datasets and we provide a comparison of different pretrained molecular encoders showing that ChemBERTa provides the best performance in our setting. These results highlight the importance of pretrained foundation models in achieving strong performance. Moreover, we expect that further improvements may come from scaling such models, which would be consistent with observations in vision-language alignment, where larger unimodal foundation models are easier to align [33].

6 Conclusion

In this work we tackle Metabolite Identification from mass spectra as a retrieval task. The most recent approaches are described in a single unified framework that encompasses the many ways to

¹<https://github.com/HassounLab/JESTR1/issues/5>

compare mass spectra and molecule representation in order to obtain a scoring function that enables retrieval. We then exploit the recent advances on multimodal alignment to tackle Molecule Retrieval as alignment of two pre-trained unimodal foundational models devoted to mass spectra and molecule representation, respectively. We analyze the behavior of our model, MSAlign, at the lens of multiple modeling choices, highlighting the relevance of joint learning of a shared representation. Additionally, we study how data split play a key role in the difficulty of the retrieval task by pointing out the risk of data shift distribution when trying to avoid data leakage. Overall, we provide a careful and reproducible comparison of recent approaches on the key benchmarks of the domain, on which MSAlign obtains the best performance. In further works we plan to examine the opportunity to even further improve the results by mixture of experts and will consider extension of this work towards *de novo* molecule elucidation.

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References

- [1] Ahmad, W., Simon, E., Chithrananda, S., Grand, G., and Ramsundar, B. (2022). ChemBERTa-2: Towards chemical foundation models. *arXiv preprint arXiv:2209.01712*.
- [2] Alseekh, S., Aharoni, A., Brotman, Y., Contrepois, K., D’Auria, J., Ewald, J., C. Ewald, J., Fraser, P. D., Giavalisco, P., Hall, R. D., Heinemann, M., Link, H., Luo, J., Neumann, S., Nielsen, J., Perez de Souza, L., Saito, K., Sauer, U., Schroeder, F. C., Schuster, S., Siuzdak, G., Skirycz, A., Sumner, L. W., Snyder, M. P., Tang, H., Tohge, T., Wang, Y., Wen, W., Wu, S., Xu, G., Zamboni, N., and Fernie, A. R. (2021). Mass spectrometry-based metabolomics: A guide for annotation, quantification and best reporting practices. *Nature Methods*, 18(7):747–756.
- [3] Bakır, G., Hofmann, T., Schölkopf, B., Smola, A. J., Taskar, B., and Vishwanathan, S. V. N., editors (2007). *Predicting Structured Data*. MIT Press, Cambridge, MA.
- [4] Bittremieux, W., Wang, M., and Dorrestein, P. C. (2022). The critical role that spectral libraries play in capturing the metabolomics community knowledge. *Metabolomics*, 18(12):94.
- [5] Bohde, M., Manjrekar, M., Wang, R., Ji, S., and Coley, C. W. (2025). DiffMS: diffusion generation of molecules conditioned on mass spectra. In *Proceedings of the 42nd International Conference on Machine Learning, ICML’25*. JMLR.org.
- [6] Brogat-Motte, L., Flamary, R., Brouard, C., Rousu, J., and d’Alché Buc, F. (2022). Learning to predict graphs with fused gromov-wasserstein barycenters. In *International Conference on Machine Learning*, pages 2321–2335. PMLR.
- [7] Brouard, C., Shen, H., Dührkop, K., d’Alché Buc, F., Böcker, S., and Rousu, J. (2016). Fast metabolite identification with input output kernel regression. *Bioinformatics*, 32(12):i28–i36.
- [8] Bushuiev, R., Bushuiev, A., de Jonge, N. F., Young, A., Kretschmer, F., Samusevich, R., Heirman, J., Wang, F., Zhang, L., Dührkop, K., et al. (2024). MassSpecGym: A benchmark for the discovery and identification of molecules. *Advances in Neural Information Processing Systems*, 37:110010–110027.
- [9] Bushuiev, R., Bushuiev, A., Samusevich, R., Brungs, C., Sivic, J., and Pluskal, T. (2025). Self-supervised learning of molecular representations from millions of tandem mass spectra using DreaMS. *Nature Biotechnology*, pages 1–11.
- [10] Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PMLR.
- [11] Chen, Y. Z., Rushing, B., and Hassoun, S. (2026). FLARE: Fine-grained learning for alignment of spectra-molecule representation enhances metabolite annotation. *bioRxiv*, pages 2026–01.
- [12] Damont, A., Darii, E., Cao, C., Legrand, A., Perret, A., Dechaumet, S., Woods, A. S., Junot, C., Tabet, J.-C., and Fenaille, F. (2025). Exploring the fragmentation of sodiated species involving covalent-bond cleavages for metabolite characterization. *Rapid Communications in Mass Spectrometry*, page e10133.
- [13] de Jonge, N., van der Hooft, J. J. J., and Probst, D. (2025). To Bin or not to Bin: Alternative Representations of Mass Spectra.

- [14] De Vijlder, T., Valkenburg, D., Lemi re, F., Romijn, E. P., Laukens, K., and Cuyckens, F. (2018). A tutorial in small molecule identification via electrospray ionization-mass spectrometry: The practical art of structural elucidation. *Mass Spectrometry Reviews*, 37(5):607–629.
- [15] De Waele, G., Wydmuch, M., Waegeman, W., et al. (2026). Small molecule retrieval from tandem mass spectrometry: what are we optimizing for? *arXiv preprint arXiv:2602.16507*.
- [16] Domingo-Almenara, X., Montenegro-Burke, J. R., Benton, H. P., and Siuzdak, G. (2018). Annotation: A Computational Solution for Streamlining Metabolomics Analysis. *Analytical Chemistry*, 90(1):480–489.
- [17] D hrkop, K., Fleischauer, M., Ludwig, M., Aksenov, A. A., Melnik, A. V., Meusel, M., Dorrestein, P. C., Rousu, J., and B cker, S. (2019). SIRIUS 4: A rapid tool for turning tandem mass spectra into metabolite structure information. *Nature Methods*, 16(4):299–302.
- [18] D hrkop, K., Nothias, L.-F., Fleischauer, M., Reher, R., Ludwig, M., Hoffmann, M. A., Petras, D., Gerwick, W. H., Rousu, J., Dorrestein, P. C., et al. (2021). Systematic classification of unknown metabolites using high-resolution fragmentation mass spectra. *Nature biotechnology*, 39(4):462–471.
- [19] D hrkop, K., Shen, H., Meusel, M., Rousu, J., and B cker, S. (2015). Searching molecular structure databases with tandem mass spectra using CSI:FingerID. *Proceedings of the National Academy of Sciences*, 112(41):12580–12585.
- [20] El Abiead, Y., Rutz, A., Zuffa, S., Amer, B., Xing, S., Brungs, C., Schmid, R., Correia, M. S. P., Caraballo-Rodr guez, A. M., Zarrinpar, A., Mannochio-Russo, H., Witting, M., Mohanty, I., Pluskal, T., Bittremieux, W., Knight, R., Patterson, A. D., van der Hooft, J. J. J., B cker, S., Dunn, W. B., Linington, R. G., Wishart, D. S., Wolfender, J.-L., Fiehn, O., Zamboni, N., and Dorrestein, P. C. (2025). Discovery of metabolites prevails amid in-source fragmentation. *Nature Metabolism*, 7(3):435–437.
- [21] El Ahmad, T., Brogat-Motte, L., Laforgue, P., and d’Alch  Buc, F. (2024). Sketch in, sketch out: Accelerating both learning and inference for structured prediction with kernels. In *International conference on artificial intelligence and statistics*, pages 109–117. PMLR.
- [22] Elser, D., Huber, F., and Gaquerel, E. (2023). Mass2SMILES: deep learning based fast prediction of structures and functional groups directly from high-resolution ms/ms spectra. *bioRxiv*, pages 2023–07.
- [23] Fan, Z., Alley, A., Ghaffari, K., and Ransom, H. W. (2020). MetFID: Artificial neural network-based compound fingerprint prediction for metabolite annotation. *Metabolomics*, 16(10):104.
- [24] Farahani, A., Voghoei, S., Rasheed, K., and Arabnia, H. R. (2021). A brief review of domain adaptation. *Advances in data science and information engineering: proceedings from ICDATA 2020 and IKE 2020*, pages 877–894.
- [25] Flamary, R., Vincent-Cuaz, C., Courty, N., Gramfort, A., Kachaiev, O., Tran, H. Q., David, L., Bonet, C., Cassereau, N., Gnassounou, T., et al. (2024). Pot python optimal transport (version 0.9. 5), 2024. [URL https://github.com/PythonOT/POT](https://github.com/PythonOT/POT), 10.
- [26] Goldman, S., Wohllwend, J., Stra ar, M., Haroush, G., Xavier, R. J., and Coley, C. W. (2023a). Annotating metabolite mass spectra with domain-inspired chemical formula transformers. *Nature Machine Intelligence*, 5(9):965–979.
- [27] Goldman, S., Xin, J., Provenzano, J., and Coley, C. W. (2023b). MIST-CF: Chemical formula inference from tandem mass spectra. *Journal of Chemical Information and Modeling*, 64(7):2421–2431.
- [28] Gupta, V., Qiang, H., Chung, H.-H., Herbst, E., and Skinnider, M. A. (2026). Comprehensive curation and harmonization of small-molecule MS/MS libraries in SpectraVerse. *Analytical Chemistry*, 98(5):3934–3943.
- [29] Han, Y., Wang, P., Yu, K., Chen, X., and Chen, L. (2025). MS-BART: Unified modeling of mass spectra and molecules for structure elucidation. *arXiv preprint arXiv:2510.20615*.
- [30] Heinonen, M., Shen, H., Zamboni, N., and Rousu, J. (2012). Metabolite identification and molecular fingerprint prediction through machine learning. *Bioinformatics*, 28(18):2333–2341.
- [31] Heirman, J. and Bittremieux, W. (2024). Reusability report: annotating metabolite mass spectra with domain-inspired chemical formula transformers. *Nature Machine Intelligence*, 6(11):1296–1302.
- [32] Hong, Y., Li, S., Ye, Y., and Tang, H. (2025). FIDDLE: a deep learning method for chemical formulas prediction from tandem mass spectra. *Nature Communications*, 16(1):11102.
- [33] Huh, M., Cheung, B., Wang, T., and Isola, P. (2024). Position: The platonic representation hypothesis. In *ICML*, pages 20617–20642.
- [34] Ji, H., Deng, H., Lu, H., and Zhang, Z. (2020). Predicting a molecular fingerprint from an electron ionization mass spectrum with deep neural networks. *Analytical chemistry*, 92(13):8649–8653.
- [35] Ji, X., Wang, Z., Gao, Z., Zheng, H., Zhang, L., Ke, G., et al. (2024). Uni-Mol2: Exploring molecular pretraining model at scale. *arXiv preprint arXiv:2406.14969*.

- [36] Kalia, A., Zhou Chen, Y., Krishnan, D., and Hassoun, S. (2025). JESTR: Joint embedding space technique for ranking candidate molecules for the annotation of untargeted metabolomics data. *Bioinformatics*, 41(7):btaf354.
- [37] Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B. A., Thiessen, P. A., Yu, B., et al. (2023). PubChem 2023 update. *Nucleic acids research*, 51(D1):D1373–D1380.
- [38] Kind, T., Tsugawa, H., Cajka, T., Ma, Y., Lai, Z., Mehta, S. S., Wohlgemuth, G., Barupal, D. K., Showalter, M. R., Arita, M., and Fiehn, O. (2018). Identification of small molecules using accurate mass MS/MS search. *Mass Spectrometry Reviews*, 37(4):513–532.
- [39] Krzakala, P., Melo, G., Laclau, C., d’Alché Buc, F., and Flamary, R. (2025). The quest for the graph level autoencoder (GRALE). *arXiv preprint arXiv:2505.22109*.
- [40] Kudriavtseva, P., Kashkinov, M., and Kertész-Farkas, A. (2021). Deep convolutional neural networks help scoring tandem mass spectrometry data in database-searching approaches. *Journal of proteome research*, 20(10):4708–4717.
- [41] Landrum, G. et al. (2013). Rdkit documentation. *Release*, 1(1-79):4.
- [42] LeCun, Y., Chopra, S., Hadsell, R., Ranzato, M., Huang, F., et al. (2006). A tutorial on energy-based learning. *Predicting structured data*, 1(0).
- [43] Litsa, E., Chenthamarakshan, V., Das, P., and Kaviraki, L. (2021). Spec2Mol: An end-to-end deep learning framework for translating ms/ms spectra to de-novo molecules. *ChemRxiv*.
- [44] Ludwig, M., Broeckling, C. D., Dorrestein, P. C., Dührkop, K., Schymanski, E. L., Böcker, S., and Nothias, L.-F. (2020). Studying charge migration fragmentation of sodiated precursor ions in collision-induced dissociation at the library scale. *Journal of the American Society for Mass Spectrometry*, 32(1):180–186.
- [45] Méndez-Lucio, O., Nicolaou, C., and Earnshaw, B. (2022). MolE: a molecular foundation model for drug discovery. *arXiv preprint arXiv:2211.02657*.
- [46] Nguyen, D. H., Nguyen, C. H., and Mamitsuka, H. (2019). Recent advances and prospects of computational methods for metabolite identification: A review with emphasis on machine learning approaches. *Briefings in Bioinformatics*, 20(6):2028–2043.
- [47] Nowozin, S. and Lampert, C. H. (2011). Structured prediction and learning in computer vision. *Foundations and Trends in Computer Graphics and Vision*, 6(3-4):3–4.
- [48] Peyré, G. and Cuturi, M. (2019). Computational optimal transport with applications to data sciences. *Foundations and Trends® in Machine Learning*, 11(5-6):355–607.
- [49] Pollmann, J., Bushuiev, R., Bushuiev, A., Pluskal, T., and Huber, F. (2026). Bridging ms2 spectra and chemical space: Advances in spectral similarity, molecular retrieval, and de novo structure discovery. *chemrxiv.15000536*.
- [50] Quinn, R. A., Melnik, A. V., Vrbanac, A., Fu, T., Patras, K. A., Christy, M. P., Bodai, Z., Belda-Ferre, P., Tripathi, A., Chung, L. K., Downes, M., Welch, R. D., Quinn, M., Humphrey, G., Panitchpakdi, M., Weldon, K. C., Aksenov, A., da Silva, R., Avila-Pacheco, J., Clish, C., Bae, S., Mallick, H., Franzosa, E. A., Lloyd-Price, J., Bussell, R., Thron, T., Nelson, A. T., Wang, M., Leszczynski, E., Vargas, F., Gauglitz, J. M., Meehan, M. J., Gentry, E., Arthur, T. D., Komor, A. C., Poulsen, O., Boland, B. S., Chang, J. T., Sandborn, W. J., Lim, M., Garg, N., Lumeng, J. C., Xavier, R. J., Kazmierczak, B. I., Jain, R., Egan, M., Rhee, K. E., Ferguson, D., Raffatellu, M., Vlamakis, H., Haddad, G. G., Siegel, D., Huttenhower, C., Mazmanian, S. K., Evans, R. M., Nizet, V., Knight, R., and Dorrestein, P. C. (2020). Global chemical effects of the microbiome include new bile-acid conjugations. *Nature*, 579(7797):123–129.
- [51] Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. (2021). Learning transferable visual models from natural language supervision. In *ICML*, pages 8748–8763.
- [52] Rakhshaninejad, M., De Waele, G., Jürgens, M., and Waegeman, W. (2026). Reliable molecular retrieval from mass spectra using conformal prediction. *bioRxiv*, pages 2026–03.
- [53] Robinson, J., Chuang, C.-Y., Sra, S., and Jegelka, S. (2020). Contrastive learning with hard negative samples. *arXiv preprint arXiv:2010.04592*.
- [54] Rong, Y., Bian, Y., Xu, T., Xie, W., Wei, Y., Huang, W., and Huang, J. (2020). Self-supervised graph transformer on large-scale molecular data. *Advances in neural information processing systems*, 33:12559–12571.
- [55] Roschmann, S., Krzakala, P., Mazelet, S., Bouniot, Q., and Akata, Z. (2026). SOTAlign: Semi-supervised alignment of unimodal vision and language models via optimal transport. *arXiv preprint arXiv:2602.23353*.
- [56] Russo, F. F., Nowatzky, Y., Jaeger, C., Parr, M. K., Benner, P., Muth, T., and Lisec, J. (2024). Machine learning methods for compound annotation in non-targeted mass spectrometry—A brief overview of fingerprinting, in silico fragmentation and de novo methods. *Rapid Communications in Mass Spectrometry*, 38(20):e9876.

- [57] Schymanski, E. L., Jeon, J., Gulde, R., Fenner, K., Ruff, M., Singer, H. P., and Hollender, J. (2014). Identifying Small Molecules via High Resolution Mass Spectrometry: Communicating Confidence. Environmental Science & Technology, 48(4):2097–2098.
- [58] Stravs, M. A., Dührkop, K., Böcker, S., and Zamboni, N. (2022). MSNovelist: de novo structure generation from mass spectra. Nature Methods, 19(7):865–870.
- [59] Thirukovalluru, R., Meng, R., Liu, Y., Su, M., Nie, P., Yavuz, S., Zhou, Y., Chen, W., Dhingra, B., et al. (2025). Breaking the batch barrier (B3) of contrastive learning via smart batch mining. arXiv preprint arXiv:2505.11293.
- [60] Vaniya, A. and Fiehn, O. (2022). Revisiting CASMI: Compound ID for 500 new unknowns, using LC-MS/MS data.
- [61] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. Advances in neural information processing systems, 30.
- [62] Vouitsis, N., Liu, Z., Gorti, S. K., Villecroze, V., Cresswell, J. C., Yu, G., Loaiza-Ganem, G., and Volkovs, M. (2024). Data-efficient multimodal fusion on a single GPU. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 27239–27251.
- [63] Wang, T. and Isola, P. (2020). Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In International conference on machine learning, pages 9929–9939. PMLR.
- [64] Wang, Y., Chen, X., Liu, L., and Hassoun, S. (2025). MADGEN: Mass-spec attends to de novo molecular generation. arXiv preprint arXiv:2501.01950.
- [65] Wishart, D. S. (2019). Metabolomics for investigating physiological and pathophysiological processes. Physiological Reviews, 99(4):1819–1875.
- [66] Wu, Z., Ramsundar, B., Feinberg, E. N., Gomes, J., Geniesse, C., Pappu, A. S., Leswing, K., and Pande, V. (2018). MoleculeNet: a benchmark for molecular machine learning. Chemical science, 9(2):513–530.
- [67] Xing, S., Shen, S., Xu, B., Li, X., and Huan, T. (2023). BUDDY: Molecular formula discovery via bottom-up MS/MS interrogation. Nature Methods, 20(6):881–890.
- [68] Xu, R. and Zhu, J. (2025). Unveiling the dark matter of the metabolome: A narrative review of bioinformatics tools for LC-HRMS-based compound annotation. Talanta, 295:128327.
- [69] Zhang, L., Yang, Q., and Agrawal, A. (2025). Assessing and learning alignment of unimodal vision and language models. In CVPR, pages 14604–14614.

A Implementation details

All SMILES strings are canonicalized and sanitized using RDKit [41]. Chemical formulas and weights are computed by explicitly accounting for implicit hydrogen atoms. The molecular masses are computed with RDKit ExactMolWt function which uses the monoisotopic atomic mass (mass of the most abundant isotope). All baselines are trained using the hyperparameters reported in their respective original works. MSAlign is trained on a single NVIDIA V100 GPU, using the AdamW optimizer with linear warmup followed by cosine decay. Following common practice, we augment the contrastive losses with a temperature parameter ϵ , for instance $\mathcal{L}_{\text{batch}}$ writes as:

$$\mathcal{L}_{\text{batch}}(\theta) = - \sum_{i=1}^B \log \frac{\exp(\rho_{\theta}(s_i, m_i)/\epsilon)}{\sum_{j=1}^B \exp(\rho_{\theta}(s_i, m_j)/\epsilon)}. \quad (9)$$

and ϵ is a trainable parameters parametrized in log scale and initialized to 0.07.

When available, we integrate metadata information (adduct and collision energy), following the basic strategy described in B.1. Note that the performance gains are modest, which suggests that this strategy could be improved in future works.

Additional hyperparameters are provided in Table 6.

The train time of all the methods that we reimplement is reported in Table 7. All DreaMS and ChemBERTa embeddings of the spectra and molecules (including candidates) are precomputed, which takes a few hours but needs to be done only once.

Table 6: MSAlign Hyperparameters.

Parameter	MassSpecGym	Spectraverse	NPLIB
<i>MLP</i>			
# hidden layers	2	2	1
# hidden dim	2048	2048	2048
# shared space dim	1024	1024	1024
dropout	0.2	0.2	0.2
layer norm	Yes	Yes	Yes
<i>Optimization</i>			
learning rate	1e−4	1e−4	1e−4
# max steps	24000	24000	16000
# warmup steps	4000	4000	4000
<i>Loss</i>			
# batch size	128	128	128
# candidates per MS	128	128	128

Table 7: Comparison of method train time on MassSpecGym. Results are reported on a Nvidia V100. Embeddings pre-computation is not included as it is sufficient to do it once.

Model	SAIL	EmbCos	JESTR	FLARE	MSAlign
Training (hours)	1.8	3.9	0.6	0.9	0.5
Finetuning (hours)	—	—	5.2	—	—
Total (hours)	1.8	3.9	5.8	0.9	0.5

B Leveraging spectra metadata

Most MS/MS datasets provide additional metadata, including adduct type, collision energy, and precursor mass. Incorporating this information could potentially improve model performance.

In this section, we explore simple strategies to integrate such metadata into MSAlign. Overall, the observed gains remain modest. This is somewhat surprising, as parameters such as collision energy are expected to play an important role in shaping fragmentation patterns. This may indicate that (i) the proposed integration strategies are insufficient, (ii) this information is already implicitly captured by the spectra representations, or (iii) performance may be approaching a plateau given the current datasets and underlying foundation models.

B.1 Basic Strategy

We consider three types of metadata: (1) precursor mass, (2) adduct type, and (3) collision energy.

Precursor Mass. The precursor mass is already incorporated in DreaMS by adding it as a peak with fixed intensity 1.1. As a result, our setting naturally accounts for this information, and we do not explore alternative encoding strategies.

Adduct. DreaMS does not explicitly encode the adduct type. To incorporate this information, we introduce a learnable embedding of dimension $d = 128$, which is concatenated to the DreaMS representation. When the information is missing we use a dedicated "unknown" embedding. As shown in Table 8, this simple approach yields a +0.8% improvement in R@1 on MassSpecGym.

Collision Energy. Collision energy is provided as a continuous value. We normalize it to the range $[0, 100]$ and encode it using sinusoidal positional encodings of dimension $d = 128$ [61]. Missing values are again represented by a dedicated (learnable) "unknown" encoding. Concatenating this encoding to the DreaMS representation leads to a +0.7% improvement in R@1 on MassSpecGym.

Overall, combining both metadata sources yields a +2.2% improvement in R@1. While non-negligible, this gain remains modest given the expected importance of these parameters.

Table 8: Metadata ablation study. We report test performance on MassSpecGym (formula split) for variants of MSAlign with or without appending metadata encodings to the spectra embedding.

Setting	R@1	R@5	R@20
w/o Adduct	52.44	72.59	87.24
w/o Collision Energy	52.38	72.00	86.45
None	51.67	71.48	86.09
Both	53.83	73.47	88.02

B.2 Finetuning expert models

As an alternative to feature-based integration of metadata, we explore leveraging adduct information through specialized models. Instead of encoding the adduct as an input feature, we train separate models for each adduct type.

Naive. In this setting, we directly train a fully separate model for each adduct type. Each model is trained only on the subset of spectra corresponding to its adduct type.

Pretraining on $[M+H]^+$. In this setting, we first pretrain a single model on a "clean" subset of the data, consisting only of spectra with the $[M+H]^+$ adduct. Then, for each adduct type, we finetune a separate model on the corresponding subset of spectra.

Pretrain on all. In this setting, we pretrain a single model on the entire dataset, which is a mixture of all adduct types. Then, for each adduct type, we finetune a separate model on the corresponding subset of spectra.

All strategies result in a set of adduct-specific “expert” models, which can be selected at inference time when the adduct is known.

In Table 9 we report the results of these different strategies. In the "Mixture of Finetuned Experts" row, we report the results that can be achieved by using the appropriate expert model for each adduct type. For the pretraining strategies, we also report the results of the pretrained model before finetuning ("Single Pretrained Model"). For completeness we also report the detailed results per adduct type.

Overall, global pretraining followed by adduct specific finetuning yields the best performance. However, the gains over a single pretrained model remain limited, indicating that most of the benefit is already captured during pretraining. In practice, this approach performs comparably to the simpler strategy of encoding the adduct as an additional feature.

Two additional observations emerge. First, pretraining exclusively on $[M+H]^+$ spectra leads to poor generalization to other adducts. Second, performance varies significantly across adduct types. This variability may reflect intrinsic differences in fragmentation patterns, as the study of adduct-specific fragmentation is itself an active area of research [12, 44], but could also be driven by dataset biases such as class imbalance or acquisition artifacts. Disentangling these factors would require a more controlled analysis and is left for future work.

Table 9: Spectraverse R@5 score on each adduct type under different training configurations.

Adduct	# Data	Pretrain on $[M+H]^+$		Pretrain on all		Train only on this adduct
		Raw	Finetuned	Raw	Finetuned	
$[M+H]^+$	233,103	—	—	58.49	58.70	56.69
$[M-H]^-$	107,867	41.31	61.22	63.19	63.90	62.20
$[M+Na]^+$	69,475	43.24	49.68	50.65	51.34	45.86
$[M+NH_4]^+$	34,889	45.22	72.35	71.93	72.83	65.91
$[M+HCOOH-H]^-$	16,136	30.07	54.95	54.66	59.93	55.54
$[M+K]^+$	9,508	32.37	35.86	43.85	44.26	30.53
$[M+CH_3COOH-H]^-$	8,456	50.93	92.48	86.38	91.08	94.36
$[M+Cl]^-$	5,958	19.35	36.91	36.55	38.35	32.61
$[M]^+$	3,405	41.37	87.06	41.37	89.65	86.20
Single Pretrained Model	All	41.13	—	59.06	—	—
Mixture of Finetuned Experts	All	—	59.08	—	60.10	57.04

C Additional Results

Keeping the MSAlign backbone fixed, we compare molecular encoders across three modalities (fingerprints, graph-based representations, and SMILES) on the retrieval task. Table 10 reports Recall@ k , showing that ChemBERTa achieves the best performance across all metrics.

Table 10: Comparison of Molecular Encoders (Fondation models and Fingerprints).

Modality	Encoder	Dim	R@1	R@5	R@20
Fingerprint	Morgan Radius 2	2048	47.93	70.26	82.01
	Morgan Radius 2 (Large)	4096	48.50	70.10	84.00
Graph	Grover	2040	46.92	69.92	86.39
	MHG-GED	1024	50.28	72.08	87.69
SMILES	SMITED	768	44.35	67.25	83.85
	ChemBERTa	768	53.83	73.11	87.10

Following the same experimental setup as Figure 3, we also compare the effect of using $\mathcal{L}_{\text{batch}}$ and $\mathcal{L}_{\text{cand}}$ on Spectraverse dataset; results are presented in Figure 5.

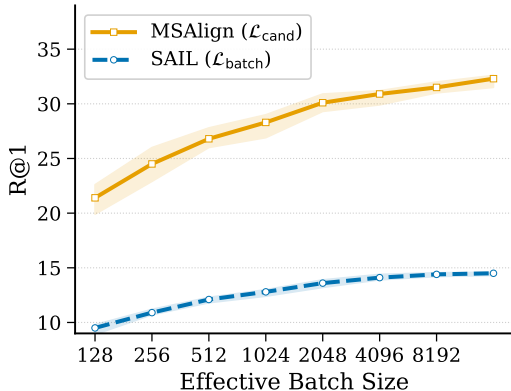


Figure 5: Effect of scaling the effective batch size on Spectraverse performances. For MSAlign the effective batchsize is $B \times K$, we fix $B = 128$ and scale the number of negatives K . For SAIL we directly scale the batchsize B .

We reproduce the encoder comparison from Table 5 on the Spectraverse dataset. As with MassSpecGym, combining ChemBERTa with DreaMS achieves the best performance across all metrics

Table 11: Comparison of different encoders E_{mol} and E_{ms} for MSAlign. Dimensions of the embeddings are indicated in parentheses. Results are reported on Spectraverse.

Encoders		Spectraverse		
Molecules (dim)	Spectra (dim)	R@1	R@5	R@20
Fingerprint (4096)	Binned (10500)	27.1	51.8	72.2
ChemBERTa (768)	Binned (10500)	26.7	54.3	74.9
Fingerprint (4096)	DreaMS (1024)	29.8	55.2	75.1
ChemBERTa (768)	DreaMS (1024)	32.3	59.1	79.6

Table 12 summarizes the statistics of the three datasets used in our experiments.

Table 12: Dataset statistics.

	NPLIB	MassSpecGym	Spectraverse
# Mol/MS pairs	10,633	231,104	488,797
# Unique molecules	8,488	28,936	44,307
Non [M+H] ⁺	25%	15%	52%

Figure 6 extends the t-SNE analysis of Figure 2 to all splits. It confirms that MCES and Murcko splits induce the largest distributional shifts between train and test sets, while Formula, InChI, and random splits yield more overlapping distributions.

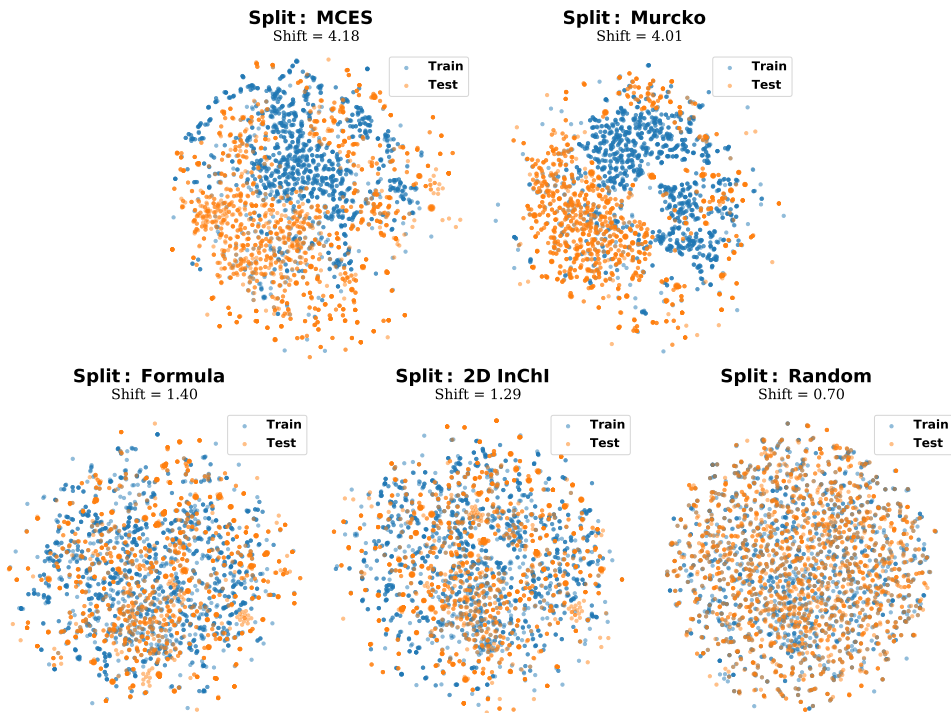


Figure 6: t-SNE visualization of all MassSpecGym splits. We sample 2000 samples from the train and test sets and visualize the embeddings (ChemBERTa and DreaMS concatenated) using 2D t-SNE.

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